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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

**Title: Cleaning and Preparing Café Sales Data with
Power Query**

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Cleaning and Preparing Café Sales Data with Power Query

Github Link: https://github.com/Amaraiah11/Business_Project_Analaysis-

Problem Statement

A retail business gathers sales information from several different channels — transactional sales data kept in Excel workbooks, customer details stored as CSV files, and product master data held either in another spreadsheet or a database. Because these sources are combined, the resulting data commonly contains gaps, repeated entries, and inconsistencies. To support dependable business reporting, management wants a single, centralized reporting solution, which means the various datasets first need to be brought into Power BI, linked correctly, cleaned up, and made ready for analysis.

Steps Followed

Data Collection

- The café sales information was obtained as one CSV file, `dirty_cafe_sales.csv`, holding 10,000 transaction rows with the fields Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.

Data Import

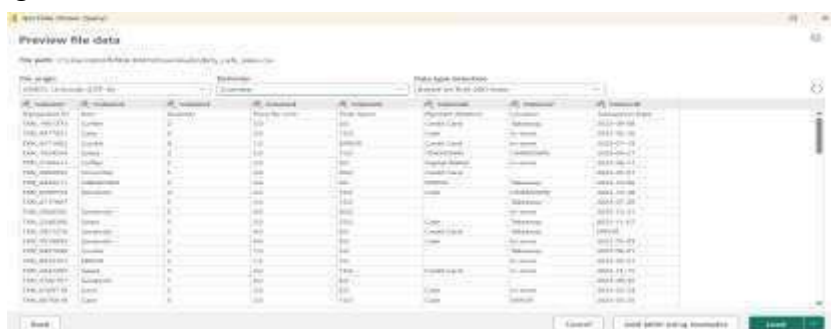
- Power BI Desktop was launched.
- The Get Data → Text/CSV option was used to browse to and open `dirty_cafe_sales.csv`.
- The file was loaded straight into the Power Query Editor so it could be cleaned and reshaped.

Data Cleaning and Transformation

Inside the Power Query Editor, a sequence of 22 individual steps was carried out, one after another, to gradually turn the messy source file into a usable dataset.

1. Get Data – Loaded `dirty_cafe_sales.csv` into the query editor.

The unprocessed café sales file was brought into Power BI through the Get Data option. Right away it was clear the file had several kinds of problems — blanks, repeated rows, error text, and inconsistent entries. It was then opened inside the Power Query Editor so the clean-up work could begin.



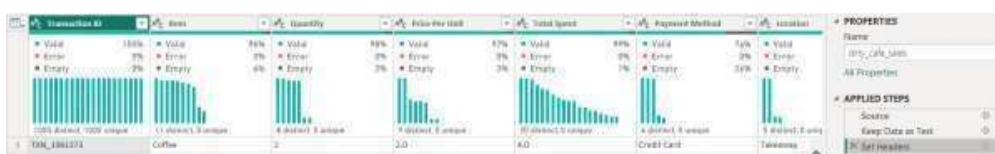
2. Keep Data as Text – Set all columns to Text initially to avoid import errors.

Every column was first switched to the Text data type. Doing this stopped Power Query from guessing at data types and silently converting messy or malformed values into the wrong format. It gave a safe starting point from which the raw values could be inspected before any real typing was applied.



3. Set Headers – Promoted the first row to become column headers.

The first row of the file (Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, Transaction Date) was promoted so it became the actual column headings. This made the structure of the table far easier to read and work with.



4. Keep Columns as Text – Reconfirmed Text type as a safe baseline before deeper cleaning.

The columns were left as Text while the first round of clean-up was carried out. Keeping them this way avoided type-conversion errors that odd entries such as ERROR, UNKNOWN, empty cells, or badly formatted numbers would otherwise have caused.

5. Fix ERROR Values – Replaced all "ERROR" placeholder entries with blanks.

Several columns — Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date — contained the literal text ERROR in place of real data. Every one of these was swapped out for a blank value so it wouldn't distort later calculations or transformations.



6. Fix UNKNOWN Values – Replaced all "UNKNOWN" placeholder entries with blanks.

A number of rows contained the word UNKNOWN, which carried no analytical value. These entries were also replaced with blanks so that later cleaning steps could deal with them consistently, alongside the other missing values.

Replace Values

Replace one value with another in the selected column(s).

Value To Find:

Replace With:

OK Cancel

7. Set Correct Data Types

With the obviously bad values cleared out, each column was finally given the data type it should have had all along. Numeric columns were converted to Number, the transaction date column was converted to Date, and the remaining descriptive columns stayed as Text.

- Quantity → Decimal Number (or Whole Number)
- Price Per Unit → Decimal Number
- Total Spent → Decimal Number
- Transaction Date → Date
- Transaction ID, Item, Payment Method, Location → Text

1.2	Quantity	1.2	Price Per Unit	1.2	Total Spent
1.2	Decimal Number	Valid	93%	Valid	95%
\$	Fixed decimal number	Error	0%	Error	0%

8. Fix Missing Total Spent

A number of rows were missing a value in the Total Spent column. Wherever Quantity and Price Per Unit were both present, the missing Total Spent figure (173 blank cells in total) was worked out by multiplying Quantity by Price Per Unit.

New column name

Total Spent Fixed

Custom column formula ⓘ

```
= if [Total Spent] = null then [Quantity] * [Price Per Unit]
  else [Total Spent]
```

9. Fix Missing Quantity

Where Quantity was blank, it was recalculated instead from Total Spent divided by Price Per Unit, provided both of those values were available. This filled in 138 missing Quantity entries.

New column name

Quantity Fixed

Custom column formula ⓘ

```
= if [Quantity] = null then [Total Spent Fixed] / [Price Per Unit] else [Quantity]
```

10. Fix Missing Unit Price

In the same way, 179 blank Price Per Unit values were derived by dividing Total Spent by Quantity whenever both figures existed for that row.

New column name

Price Fixed

Custom column formula ⓘ

```
= if [Price Per Unit] = null then [Total Spent Fixed] /
[Quantity Fixed] else [Price Per Unit]
```

11. Remove Old Columns

Once the corrected figures had been calculated, the original columns they replaced were no longer needed and were removed. This kept the table tidy and left only the fields required for the final dataset.

12. Fix Column Names

Column names were reviewed and, where needed, renamed so they were clear, consistent, and self-explanatory — a small change that makes any later analysis much easier to follow.

13. Check Data Types

After the new calculated columns were added, the data types were checked once more to make sure Quantity, Price Per Unit, Total Spent, and Transaction Date were all still stored correctly.

Transaction ID	Item	Payment Method	Location	Transaction Date
1.2	Total Spent	1.3	Quantity	1.2
	Price Per Unit			

14. Fix Missing Items

333 rows had no value in the Item column. These blanks were replaced with the placeholder label "Unknown Item" so the rows remained usable rather than being discarded.

15. Fix Missing Payment Method

2,579 rows were missing a Payment Method. These were also filled in with a placeholder label, "Unknown," which kept the payment information complete and consistent across the whole dataset.

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

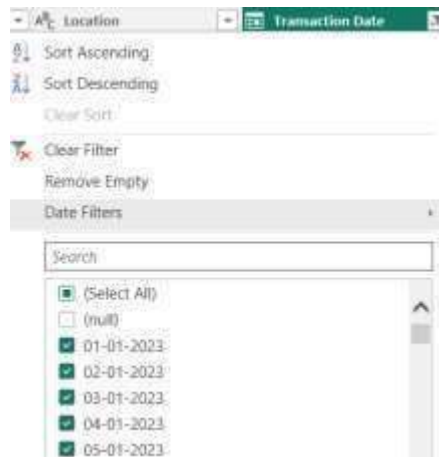
Unknown Item

16. Fix Missing Location

The Location field had the largest number of gaps in the whole file — 3,265 blank entries. These were addressed during the clean-up so that incomplete location data wouldn't get in the way of later analysis.

17. Remove Missing Dates

159 rows had no Transaction Date and were removed outright. Since the date tells you when a sale actually happened, rows without one can't be trusted for any date-based reporting.



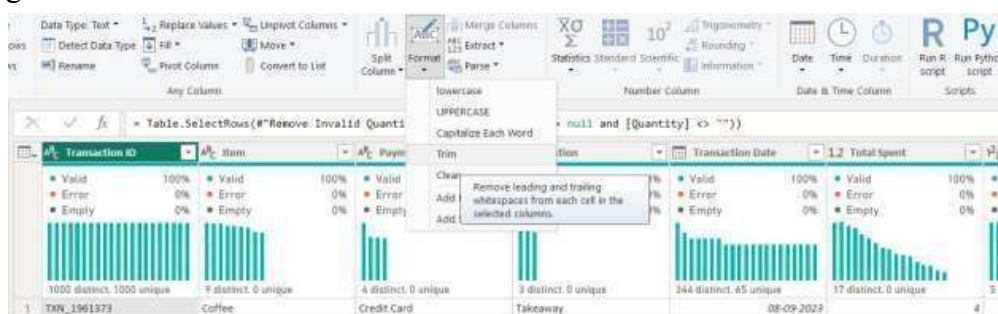
18. Remove Duplicate Records

The table was checked for fully duplicated rows. None turned up, which confirmed that every Transaction ID was already unique. Checking for duplicates matters because it stops any single café transaction from being counted more than once in the figures.



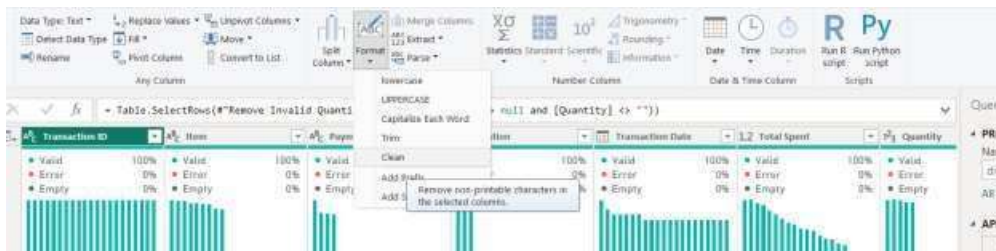
19. Fix Extra Spaces

Leading and trailing spaces were stripped from every text column using the Trim operation. This meant entries like " Coffee" and "Coffee " were now recognized as the same value instead of being treated as different items.



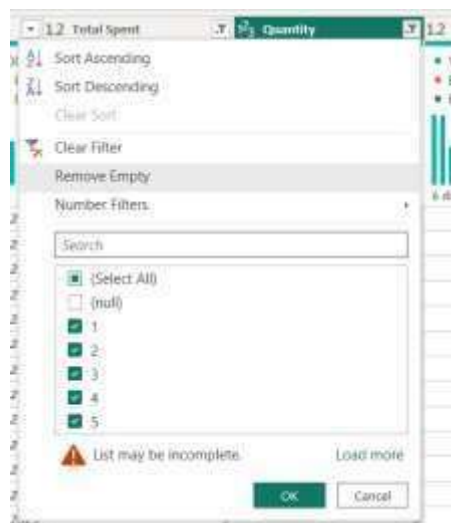
20. Fixed and Cleaned

Non-printable and hidden characters were stripped out of the text fields with the Clean operation. This step raises the overall quality of the text data and stops invisible characters from creating false mismatches between otherwise identical entries.



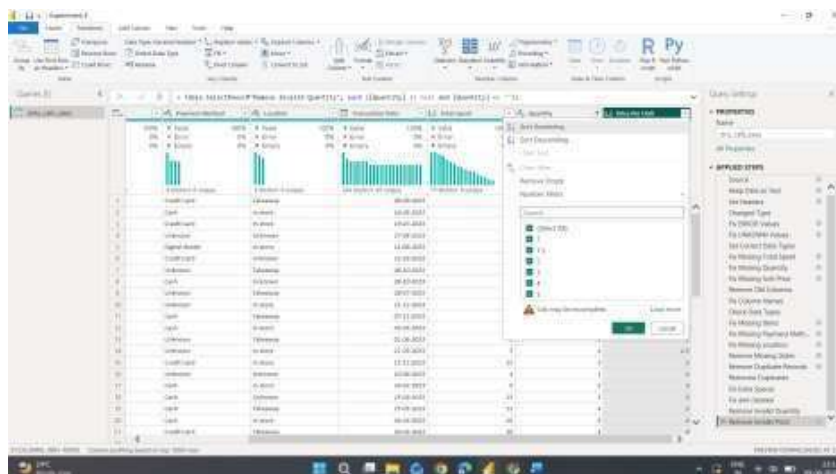
21. Remove Invalid Quantity

The Quantity column was scanned for anything that didn't make sense — negative numbers or other unsuitable values — and those rows were dropped, since a genuine café sale can only ever have a positive quantity.



22. Remove Invalid Price

Finally, the Price Per Unit column was checked for negative or otherwise invalid figures. Rows with unrealistic prices were removed so that the finished dataset only contained sensible, trustworthy pricing information.



Taken together, this sequence of steps eliminated duplicate rows, resolved missing values (either by calculation, by inserting a placeholder label, or by removing the row entirely), gave columns clearer names, and converted every field to its proper data type.

Outcome

Once all 22 transformation steps had been applied in Power Query, the originally messy café sales file was fully cleaned and organized. Error text, UNKNOWN placeholders, missing values, duplicate rows, wrong data types, stray spaces, hidden characters, and invalid quantity or price entries were all identified and resolved. Wherever possible, missing Total Spent, Quantity, and Price Per Unit values were also recalculated rather than simply discarded.

The result is a dataset that is noticeably more accurate, consistent, and dependable than the original raw export. Columns now carry clear, sensible names and correct formatting, and records that could not be trusted have been removed. The cleaned café sales table is now properly structured and ready to support further analysis and reporting in Power BI.

Conclusion

This exercise walked through a full data-preparation workflow in Power BI and Power Query, starting from a raw café sales file and ending with a clean, analysis-ready dataset. The source data was first pulled into Power Query using the Get Data option and reviewed for quality problems, which included missing values, ERROR and UNKNOWN placeholder text, duplicate rows, incorrect data types, extra whitespace, hidden characters, and invalid quantity or price entries.

Power Query's step-based transformation tools were then applied one after another to work through these problems systematically. Missing Total Spent, Quantity, and Price Per Unit values were reconstructed wherever the surrounding data allowed it, unusable or duplicate rows were removed, and text fields were trimmed and standardized. Correct data types were applied throughout so that every numeric, text, and date field would behave properly in later calculations.

By the end of this process, the original dirty café sales file had been turned into a clean, wellorganized, and trustworthy dataset. The exercise illustrates how much data cleaning contributes to the accuracy and consistency of information before it's used for analysis. The finished dataset is now properly structured and ready to support further work in Power BI, giving future calculations, reports, and business decisions a solid, reliable foundation.

Business Insights

- A clean dataset helps the café keep accurate, trustworthy sales records.
- Missing Total Spent, Quantity, and Unit Price figures can often be reconstructed from the sales data that is available.
- Removing duplicate rows stops the same sale from being counted more than once.
- Correcting ERROR and UNKNOWN entries makes the whole dataset more reliable.
- Dropping invalid or negative Quantity and Price values keeps the sales records realistic.
- Standardizing Item, Payment Method, and Location entries makes the data more consistent overall.
- Removing records with no Transaction Date protects the accuracy of the sales history.
- Correct data types make future sales analysis considerably easier.
- A clean dataset gives the café a better basis for decisions about sales, pricing, inventory, and customer preferences.

- Overall, Power Query is what turns messy café sales data into a clean, reliable dataset that's ready for analysis.